



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 206: ELECTRICAL MACHINES I

2ND SEMESTER, 2022/2023

TIME: 2 Hours

INSTRUCTIONS:

Answer Question ONE and any other TWO questions

Question 1 (30 marks)

- a. With aid of diagram and expressions differentiate between back pitch , from pitch and resultant pitch for a lap wound dc machine.. (5 marks)
- b. Figure 1 show a magnetic circuit made of different materials. Section A is wound with 500 turns. The materials used to make the entire magnetic circuit has uniform cross section area of 500 mm squared. The lengths of section .A is 150 mm and relative permeability of 800. The lengths of section B is 400 mm and relative permeability of 1000. The lengths of section C is 80 mm and relative permeability of 700. The lengths of section D is 100 mm and relative permeability of 600. Determine.
- (i) Total reluctance of the circuit
 - (ii) Determine magneto motive force required to set up a flux 1.0 mWb in the limb with the coil.
 - iii) Calculate the current flowing in the coil
 - iv) determine flux in Section B
 - v) The flux in Section C (10 marks)

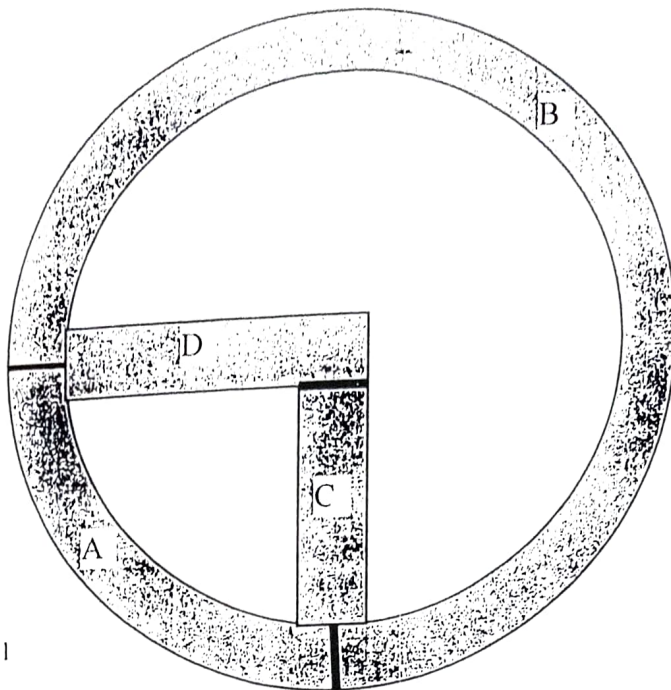


Fig. 1

- c. The temperature rise of a loaded transformer is 25°C after one hour and 37.5°C after two hours. When disconnected, the temperature falls from the final steady state temperature to 40°C in 2.8 hours. Taking the ambient temperature to be 30°C . Calculate:
- Heating time constant
 - Final steady full load temperature rise.
 - Cooling time constant.
- (9 marks)
- d. An 8 pole dc shunt generator with 778 wave wound armature conductors and running at 500 rpm, supplies a load of 12.5 Ohms resistance at a terminal voltage of 250 volts. The armature resistance is 0.24 Ohms and the field resistance is 250 Ohms. Calculate:
- The armature current
 - The induced emf
 - The flux per pole.
- (6 marks)

Question 2 (20 marks)

- a. Define the following terms:
- Magnetizing force
 - Magnetic field density
- (3 marks)
- b. Differentiate between progressive and retrogressive lap wound windings. (3 marks)
- c. An eight pole lap wound dc machine has 2 armature slots per pole and 2 conductors per slot.
- Develop the winding table
 - Prepare the development diagram
 - Determine the number of parallel paths
 - Indicate the position of the brushes.
- (8 marks)
- d. A 400/200V single phase transformer is supplying a load of 50A at a power factor of 0.8 lagging. The no load current is 2A at 0.208 power factor lagging. Calculate the primary current and the primary power factor. (6 marks)

Question 3 (20 marks)

- State Faraday's laws of electromagnetic induction. (2 marks)
- Figure 2 is a hysteresis loop diagram.
 - Briefly explain the behavior of curve from O to P.
 - Describe point Y.
 - For the core of an transformer, would you use core material with a wide hysteresis loop or with a narrow hysteresis loop. (6 marks)

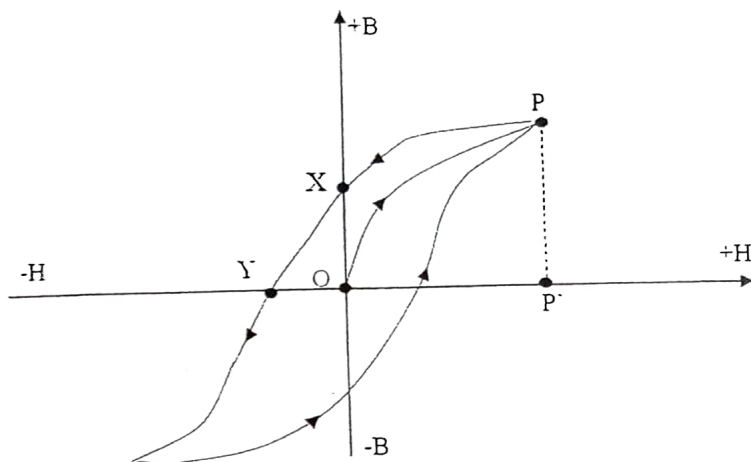


Fig. 2

- Calculate the resistance steps for the starter of a 250 Volt dc shunt motor having armature resistance of 0.125 Ohm and a full load current of 150 amperes. The motor is to start against a full load and the maximum current is not to exceed 200 amperes. (6 marks)
- With aid of labelled diagrams and equations, describe transformer open circuit and short circuit tests and their purpose. (6 marks)

Question 4 (20 marks)

- An Engineer in KACHO company was requested to design a high voltage low current dc motor. With reasons, explain whether he would choose wave or lap winding. (3 marks)
- A coil carrying a 7.16 ampere current is wound on a wooden ring which has a cross section area of 200 mm squared. The wooden ring has a mean circumference of 900mm. determine:
 - The opposition the ring offers to the flow of flux.
 - The number of coil turns. (4 marks)
- A 400 kVA transformer has primary winding resistance of 0.5 Ohm and secondary winding resistance of 0.001 Ohm. The iron loss is 2.5kW and the primary and secondary voltages are 5kV and 320V respectively. If the power factor of the load is 0.85, determine:
 - Transformer efficiency at full load
 - Transformer efficiency at half load.
 - Load at which maximum efficiency occurs.
 - Maximum efficiency. (8 marks)
- A shunt DC motor is rated at 240V and runs at 1000 rev/min at no load. It is required to increase the speed of the motor to 1250rpm while drawing a current of 40A from the mains.

The armature resistance and field resistance are 0.25 and $150\ \Omega$ respectively. Determine the resistance to be placed in series with the field so that the motor runs at this new speed. Assume the pole flux to be proportional to the field current and neglect rotational losses.
(5 marks)

Question 5 (20 marks)

- a. Explain why transformers are rated in KVA. (2 marks)
- b. Briefly discuss the TWO drawbacks of brush shifting as a way of improving commutation. (4 marks)
- c. A dc generator running at 30 rev/sec generates an emf of 200 volts . Determine the percentage increase in the flux per pole required to generate 250 volts at 20 rev/sec . (4 marks)
- d. The copper losses in transformer varies with load. The iron losses are constant from no load to full load. With aid of diagram, explain why core losses are constant irrespective of the load. (4 marks)
- e. Two single phase transformers X and Y are operating in parallel and supplying a common load of 1000KVA at 0.8 power factor. Transformer X is rated 800KVA and its equivalent impedance referred to the secondary windings is $3+j5\text{ Ohms}$. Transformer Y is rated 600KVA and its equivalent impedance referred to the secondary windings is $2+j4\text{ Ohms}$. Determine the loading of each transformer. (6 marks)



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF ENGINEERING

ECU 202: ENGINEERING MATHEMATICS IV

2ND SEMESTER 2022/2023

TIME: 2 HRS

INSTRUCTIONS

Answer Question One and any other Two questions

QUESTION ONE (30 MARKS)

a) Determine order and linearity of the given differential equations

i) $(1 + y^2) \frac{d^2y}{dt^2} + t \frac{dy}{dx} + y = e^t$ (1 mark)

ii) $\left(\frac{d^3y}{dx^3}\right) + \left(\frac{d^2y}{dx^2}\right)^4 + \ln y = 0$ (1 mark)

b) Solve the first order linear differential equation $\frac{dy}{dx} - \frac{5}{x}y = -5x^2$ (3 marks)

c) Find the differential equation corresponding to $y = ae^{4x} + be^{-x}$, where a and b are arbitrary constants. (4 marks)

d) Find the particular solution to the initial-value problem $y'' + 4y = 0$, $y(0) = 0$, $y'(0) = 1$, if the general solution is $y(x) = C_1 \sin 2x + C_2 \cos 2x$. (3 marks)

e) Solve $(D^2 - 3D - 4)y = 0$ (3 marks)

f) Solve $\sin^{-1} x \, dy + \frac{y}{\sqrt{1-x^2}} \, dx = 0$, given that $y = 2$ when $x = \frac{1}{2}$ (5 marks)

g) Find the orthogonal trajectories for the family of curves whose equation is $y = Cx^3$ (5 marks)

h) An object with mass 10 kg is dropped from a height of 200 m. Given that its drag coefficient is $k = 2.5 \text{ N/(m/s)}$, after how many seconds does the object hit the ground? (5 marks)

QUESTION TWO (20 MARKS)

- a) Solve the homogenous differential equation $\frac{dy}{dx} = \frac{x-y}{x+y}$ (6 marks)
- b) Solve $\frac{dy}{dx} = \frac{x+2y+1}{2x+4y+3}$ (6 marks)
- c) Use the method of variation of parameters to solve
 $(x^2 - 1)y'' - 2xy' + 2y = (x^2 - 1)^2$ (8 marks)

QUESTION THREE (20 MARKS)

- a) Test the following equation for exactness hence solve.
 $x(1 - \sin y)dy = (\cos x - \cos y - y)dx$ (5 marks)
- b) Solve $\frac{d^2x}{dt^2} - 3\frac{dx}{dy} + 2x = 0$, given that $t = 0$ when $x = 0$ and $\frac{dx}{dt} = 1$ (5 marks)
- c) Solve $(D^4 + 2D^3 - 7D^2 + 2D + 70)y = 0$, if one of the roots is $-3 + i$ (6 marks)
- d) Cabalt-60 is a radioactive element that is used in medical radiology. It has a half-life of 5.3 years. Suppose that an initial sample of Cobalt-60 has a mass of 100 grams. How long will it take for 90% of the sample to decay? (4 marks)

QUESTION FOUR (20 MARKS)

- a) Solve the initial Value problem $y'' - 4y' - 12y = 3e^{5t}$, $y(0) = \frac{18}{7}$, $y'(0) = -\frac{1}{7}$ (7 Marks)
- b) Find the general solution to the Bernoulli differential equation
 $\frac{dy}{dx} + xy = xe^{-x^2}y^{-3}$ (5 Marks)
- c) The size of a certain bacterial colony increases at a rate proportional to the size of the colony. Suppose the colony occupies an area of 0.25 square centimeters initially, and after 8 hours it occupied an area of 0.35 square centimeters.
- Estimate the size of the colony t hours after the initial measurement. (4 marks)
 - What is the expected size of the colony after 12 hours? (4 marks)
 - Find the doubling time of the colony. (4 marks)

QUESTION FIVE (20 MARKS)

a) Consider the linear system of differential equations

$$y_1' = y_1 + 2y_2 + 2e^{4t}$$

$$y_2' = 2y_1 + y_2 + e^{4t}$$

- i. Write the system in matrix form and conclude that every initial value problem for the system has a unique solution on $(-\infty, \infty)$ (2 Marks)
- ii. Verify that

$$y = \frac{1}{5} \begin{bmatrix} 8 \\ 7 \end{bmatrix} e^{4t} + c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t}$$

is a solution of the above linear system of differential equations for all values of the constants c_1 and c_2 (3 Marks)

- iii. Find the solution of the initial value problem

$$y' = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} y + \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{4t} ; \quad y(0) = \frac{1}{5} \begin{bmatrix} 3 \\ 22 \end{bmatrix} \quad (3 \text{ marks})$$

b) A ceramic insulator is baked at 400°C and cooled in a room in which the temperature is 25°C .

After 4 minutes the temperature of the insulator is 200°C . What is its temperature after 8 minutes? (6 Marks)

c) If $x^k y^m$ is an integrating factor for $\frac{dy}{dx} = \frac{-(y+x^4 y^2)}{x}$, find the values of k and m , hence solve the differential equation. (6 Marks)



ECU202: ENGINEERING MATHEMATICS VI

DATE: 15th February, 2023

CAT 1

Instructions

Answer ALL the questions

1. Determine the order, degree and linearity of the following equations

a. $y''' + \sqrt{x}y + x^2(y'')^4$ (1 mark)

b. $17y^{(4)} - t^6y^{(2)} - 4y^5 = 3 \cos t$ (1 mark)

2. Determine C_1 and C_2 so that $y(x) = C_1e^{2x} + C_2e^x + 2\sin x$ will satisfy the condition $y(0) = 0$ and $y'(0) = 1$ (5 marks)

3. Find the differential equations on a plane representing all curves of radius 5. (5 marks)

4. Determine if the differential equation $\frac{dy}{dx} = \frac{x^2+y}{x^3}$ is homogenous hence state the order. (3 marks)

5. Determine if the differential equation $ydx - xdy = 0$ is exact. (2 marks)

6. Solve the exact differential equation $\frac{dy}{dx} = \frac{2+ye^{xy}}{2y-xe^{xy}}$ (5 marks)

7. Solve the homogenous differential equation $\frac{dy}{dx} = \frac{3xy}{x^2-y^2}$ (4 marks)

8. Solve $\frac{dy}{dx} - \frac{5}{x}y = -5x^2$ (4 marks)

"Study as if you were going to live forever; live as if you were going to die tomorrow."
~Maria Mitchell



ECU202: ENGINEERING MATHEMATICS VI

DATE: 16th March, 2023

CAT II

Instructions

Answer ALL the questions

1. Solve the following equations

a. $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} = \cos x, y(0) = y'(0) = 1$ (8 mark)

b. $3 \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 2y = 0$ (3 mark)

2. Use the method of undetermined coefficients to solve $y'' + 4y' + 8y = e^{-2x}$ (10 marks)

3. A body at a temperature of 50°F is placed outdoors where the temperature is 100°F. If after 5 minutes the temperature of the body is 60°F. Find

a. The expression for the temperature of the body at any time t (5 marks)

b. Time taken for the body to reach a temperature of 75°F (2 marks)

c. The temperature of the body after 20 minutes (2 marks)

"Strive for perfection in everything you do. Take the best that exists and make it better. When it does not exist, design it."

—Sir Henry Royce, English engineer and car designer

$$\int_0^1 \tan x =$$

$$e^{-2x}$$
$$e^{Ax+A_0}$$

$$\sin x = \cos x$$

$$\tan x \int \frac{\cos x}{\sin x}$$

EEE206 CAT2 50 minutes

- a. List THREE differences between LAP and WAVE wound dc machine. (5 marks)
- b. Sketch a well labelled cross section of a typical dc machine. (5 marks)
- c. Derive the expression of generated voltage of a dc machine. (5 marks)
- d. A shunt generator delivers 195 A at terminal p.d of 250 V, the armature resistance and shunt field resistance are $0.02\ \Omega$, and $50\ \Omega$ respectively. The iron and friction losses equal 950 W. Find (5 marks)
- iv. E.M.F generated
 - v. Copper losses
 - vi. Output of the prime mover
- e. separately-excited generator develops a no-load e.m.f. of 150 V at an armature speed of 20 rev/s and a flux per pole of 0.10 Wb. Determine the generated e.m.f. when (i) the speed increases to 25 rev/s and the pole flux remains unchanged, (ii) the speed remains at 20 rev/s and the pole flux is decreased to 0.08 Wb (5 marks)
- f. A two pole lap wound dc machine has 4 armature slots and 2 conductors per slot.
- i. Develop the winding table
 - ii. Prepare the development diagram
 - iii. Determine the number of parallel paths
 - iv. Indicate the position of the brushes. (5 marks)

1-6 6-3
3-8 8-5
5-2 2-7
7-4 4-1



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL ENGINEERING

EEE 209: FLUID MECHANICS

2ND SEMESTER 2022/2023

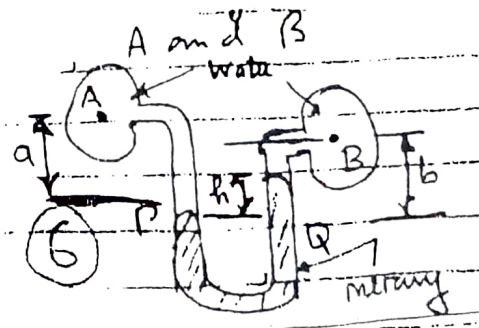
TIME: 2 HOURS

INSTRUCTIONS:

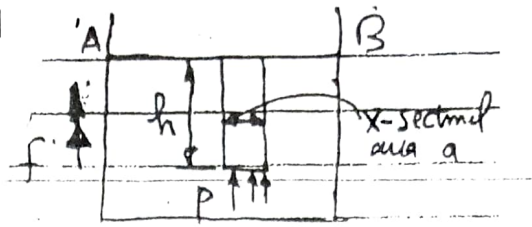
- This paper contains FIVE Questions.
- Question ONE is Compulsory and carries 30 Marks.
- The remaining FOUR Questions carry 20 Marks each.
- Attempt Question ONE and any TWO of the remaining FOUR Questions

QUESTION ONE (Compulsory Question) - 30 Marks.

- a) Concisely explain the meaning of the following terms.
- | | |
|----------------------|-------------|
| i. Absolute pressure | (2 marks) |
| ii. Vapour pressure | (2 marks) |
| iii. Surface tension | (2 marks) |
| iv. Compressibility | (2 marks) |
- b) Show that water is practically incompressible compared to air. (3 marks)
- c) i) State Pascal's law (2 marks)
- ii) Verify Pascal's Law in a liquid (5 marks)
- d) Figure 1 shows a mercury manometer for measuring pressure difference between the liquids A and B.
- If $a = 1.3 \text{ m}$ $b = 0.5 \text{ m}$ and $h = 0.15 \text{ m}$ Calculate the pressure difference between A and B.
- The specific gravity of mercury = 13.6



- e) i) A vessel containing a liquid of mass density $\rho \text{ kg/m}^3$ is given a constant acceleration of $f \text{ m/sec}^2$ in a vertical upward direction. Considering a vertical prism indicated in Fig 1, show that $P = \rho gh \left[1 + \frac{f}{g} \right]$ (3 marks)



- ii) A vessel containing water is given a constant acceleration of 3.95 m/sec^2 in an upward vertical direction.

The diameter of the cylindrical vessel is 1.5 m with a depth of water at 0.9 m . (3 marks)

Calculate the force at the bottom of the vessel while it is being accelerated.

Total = 30 marks)

QUESTION TWO (20 MARKS)

- a) A circular lamina of radius r is tangentially and vertically submerged in water. The top edge of the perimeter of the lamina just touches the surface of water. Derive an expression for the
- total force acting on one face of the lamina in terms of r , mass density and acceleration due to gravity in m/sec^2 .
 - vertical depth of the center of pressure vertically below the free surface in terms of r .
- b) A circular lamina of diameter 1.4 m is vertically submerged in water such that its top edge is tangential to the water surface.

Calculate the (i) total force acting on one face of the lamina.

- vertical depth of the centre of pressure of the plate below the water surface.

Total = 20 marks

QUESTION THREE:

- a) Derive an expression for the theoretical discharge through a horizontal venturimeter with a U-tube mercury manometer.

Work from first principles and explaining the meaning of all symbols used. (10 marks)

- b) The discharge in a 600mm diameter horizontal water main measured by means of a venturimeter with a throat diameter 300mm. The difference in pressure between the pipe and the throat corresponds to 250mm of mercury.
- Calculate the discharge in m^3/sec through the meter. Assume; the coefficient of discharge through meter is 0.99
- (10 marks)

Total = 20 marks

: Specific gravity of mercury = 13.6

: acceleration due to gravity = 9.81 m/sec^2

QUESTION FOUR: (20 MARKS)

- a) Distinguish between turbines and (pumps, fans etc) (4 marks)
- b) Using usual notations, explaining the meaning of all symbols and working from styles derive an expression for the.
- i) work due per unit weight (4 marks)
- ii) hydraulic efficiency for an inward flow reaction turbine (4 marks)
- c) An inward flow reaction turbine operating under a head of 10 metres having external and internal diameters 0.5 m and 0.3 m respectively, discharges $1.05 \text{ m}^3/\text{sec}$ of water. Water enters and leaves the vains radially. Therefore $V_{r1} = f_1$ and $V_2 = f_2$, using usual notation; Calculate the Hydraulic Efficiency of the turbine. (8 marks)

Total = 20 marks

QUESTION FIVE (20 MARKS)

- a) Working from first principles, stating assumptions made and explaining the meaning of all symbols used derive Bernoulli's Equation for an incompressible, inviscid fluid by momentum consideration method. (8 marks)
- b) Oil of mass density 900 kg/m^3 flows through a horizontal pipe which reduces smoothly from 75mm to 50mm diameter. The pressure at these points are 70 kN/m^2 and 40 kN/m^2 respectively.
- Calculate the
- (i) velocity in the larger diameter in m/sec . (6 marks)
- (ii) flow rate in kg/sec . (6 marks)

Total = 20 marks

Question Four

c) An inward flow reaction turbine operating under a head of 12 metres having external and internal diameters 0.5m and 0.3m respectively discharges 1.05m³/sec of water while running at 375rpm

Water enters and leaves the vanes radially therefore $V_{r1}=f_1$ and $V_2=f_2$ using usual notations.

Calculate the i) Power Output of the turbine (3marks)

ii) Hydraulic efficiency of the turbine (5 marks)

Total = 20 marks

KENYATTA UNIVERSITY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC
ENGINEERING

CAT 1

Monday 27/02/2023

EEE 209 FLUID MECHANICS.

Attempt the following question.

QUESTION: 20 marks

(a) A flat plate is inclined at θ° to the horizontal and is completely immersed in water. Derive an expression for

(i) total force acting on its one face in terms of its surface area, mass density of water, acceleration due to gravity and the vertical depth of its centre of area below the free surface.

(ii) the vertical depth of its centre of pressure below the free surface in terms of its radius of gyration, ~~area~~ depth of its centre of area vertically below the free surface, and angle of inclination to the horizontal.

(b) A circular lamina of diameter 200 mm is inclined and completely immersed in water such that the upper perimeter is 0.7 m ~~vertically~~ vertically below the free surface and the lower perimeter is 1.6 m vertically below the free surface. Calculate the :-

(i) total force acting on its face (4)

(ii) vertical depth of the centre of pressure below the free surface. (6)

Total = 20 marks

KENYATTA UNIVERSITY

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

CATS II

Monday

03/04/2023

EEE 209 FLUID MECHANICS

Attempt the following question

QUESTION (20 marks)

(a) Show that the power output by a Pelton wheel is given by $P = \rho Q [V_1 - \omega r] [1 - k \cos \theta] \omega r$

where ρ = mass density of water, kg/m^3

Q = discharge in m^3/sec

ω = angular rotation of the wheel in radians/sec

V_1 = absolute velocity of incoming water from the nozzle (m/sec)

(10)

θ = angle through which the water is turned at the bucket in degrees.

$k = \frac{V_{r2}}{V_{r1}}$ [using usual notation]

r = effective radius of the wheel.

(b) A Pelton wheel under a head of 140m, with a wheel diameter 2m, the deflecting angle of the buckets 165° , having a nozzle diameter 120mm runs at a speed of 250 rpm.

~~10~~ Determine the

(i) power output (3)

(ii) hydraulic efficiency (3)

$k=1$

$C_v=0.97$

Total = 20 marks



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL & ELECTRONIC ENGINEERING

EEE 208: ELECTRICAL MEASUREMENTS

2nd SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS:

Instructions: *Attempt question one and any other two questions.*

Show all working in a neat and legible manner.

Every question must start on a fresh page and be clearly labeled.

Question ONE (30 Marks) Compulsory

- a) Briefly explain the following classification of standards as used in measurements.
- i. International standard
 - ii. Primary standard
 - iii. Secondary standard
 - iv. Working standard **(6 marks)**
- b) Differentiate between Current transformer and Potential transformer. **(6 marks)**
- c) Briefly explain scenarios where measurement instruments need calibration **(6 marks)**
- d) Differentiate between the following terms
- i. Precision and accuracy
 - ii. Threshold and resolution **(2 marks)**
- e) The waveforms shown in Figure 1(e) were observed on the screen of an oscilloscope:

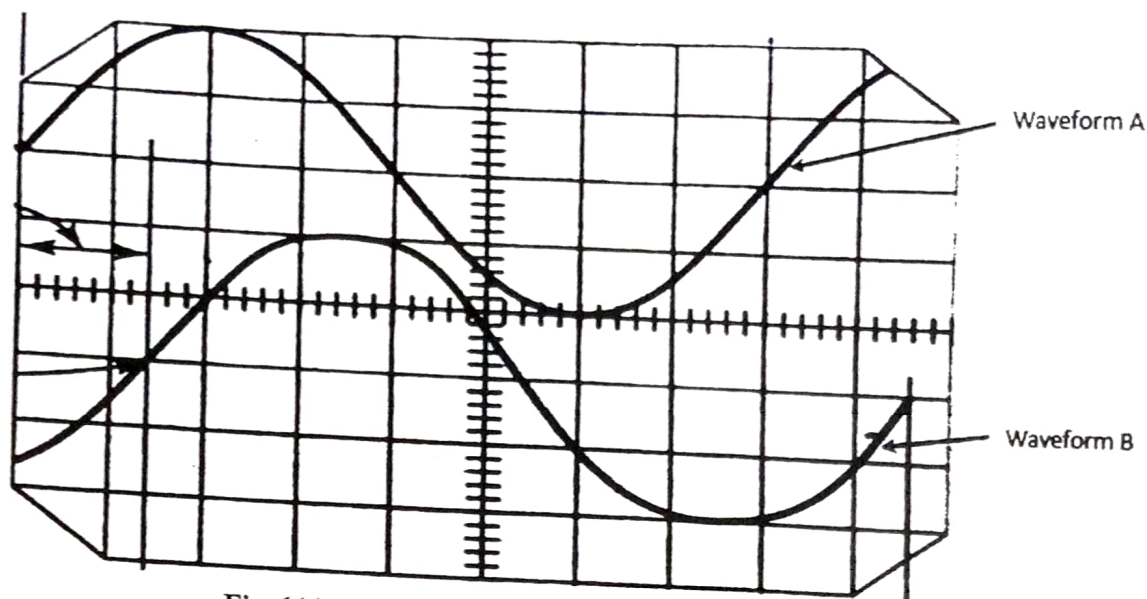


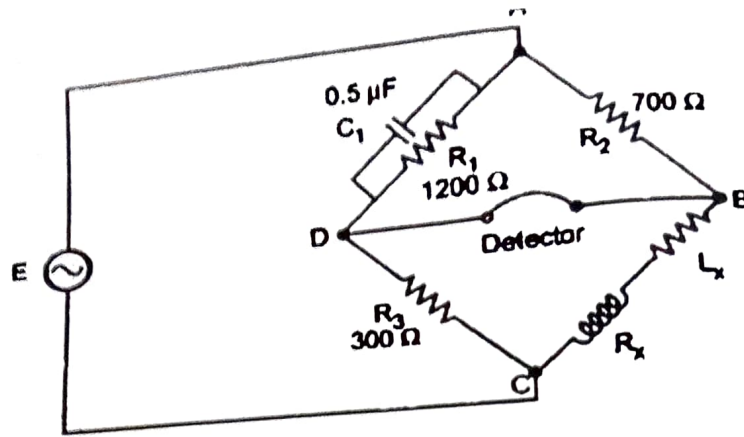
Fig. 1(e): Sampled waveforms from a CRO.

Given that the vertical attenuation is set to 3V/div and the time/div control set at $3\mu\text{S}/\text{div}$, Determine:

- i. The peak to peak amplitude of Waveform A. (2 marks)
 - ii. The frequency of Waveform A. (2 marks)
 - iii. The rise time of the Waveform B. (2 marks)
 - iv. The phase difference between the two waveforms. (2 marks)
- f) A Wheatstone bridge requires a change of 7 ohms in the unknown arm of the bridge to produce a change in deflection of 3 mm of the galvanometer. Determine the sensitivity of the bridge. (2 marks)

Question TWO (20 Marks)

- a) Differentiate between the following terms as applied in electrical measurements:
- i. Null type and deflection type instruments.
 - ii. Active and passive instruments.
 - iii. Smart and non-smart instruments. (6 marks)
- b) Explain the principle of operation and typical applications of the following transducers
- i. Magnetic reluctance
 - ii. Hall effect (4 marks)
- c) For the bridge circuit below, derive the expressions for determining the unknown resistance R_x and inductance L_x . Hence determine these unknown values given that $R_1 = 1200\ \Omega$, $R_2 = 700\ \Omega$, $R_3 = 300\ \Omega$ and $C_1 = 0.5\ \mu\text{F}$ (6 marks)



- d) The resistor R_1 and R_2 are connected in parallel. The value of the resistance are $R_1 = 200 \pm 0.2\Omega$ and $R_2 = 80 \pm 0.05\Omega$. Determine the uncertainty in the combined resistance (4 marks)

Question THREE (20 Marks)

- a) Discuss the advantages of instrument transformers as compared to shunts and multipliers for extension of instrument range. (6 marks)
- b) A current transformer has two turns on the primary winding and a secondary winding of 260 turns. The secondary winding is connected to an ammeter with a resistance of 0.2Ω . The resistance of the secondary winding is 0.3Ω . If the current in the primary winding is 500A, determine:
- The reading on the ammeter,
 - The potential difference across the ammeter,
 - The total load in VA on the secondary.
- (6 marks)
- c) What is an electrical transducer? Briefly explain, with the help of a diagram, the operation of an inductive transducer. (8 marks)

Question FOUR (20 Marks)

- a) A moving-coil instrument gives the full-scale deflection of 10 mA when the potential difference across its terminals is 100 mV. Calculate
- the shunt resistance for a full-scale deflection corresponding to 100 A,
 - the series resistance for full scale reading with 1000 V.
- Calculate the power dissipation in each case. (8 marks)

- b) A resistor has a nominal value of $10\Omega \pm 1\%$. A voltage is applied and the power consumed is calculated using the expression $P = E^2/R$. Calculate the percentage expected uncertainty in measurement of power when the measured value of $E = 100V \pm 1\%$ (8 marks)
- c) An ammeter has a full scale deflection of 100 mA and a resistance of 50Ω . The ammeter is used to measure the current in the load of resistance 500Ω when the supply voltage is 10 V. Calculate
- (i) The ammeter reading expected (neglecting its resistance),
 - (ii) The actual current in the circuit,
 - (iii) The power dissipated in the ammeter and
 - (iv) The power dissipated in the load.
- (4 marks)

Question FIVE (20 Marks)

- a) Draw the block diagram of a general purpose CRO and explain the purpose of the following controls: (14 marks)
- i. Grid
 - ii. Vertical and horizontal deflection systems.
 - iii. Time base generator
 - iv. Anodes
- d) Explain any **FOUR** methods used to measure medium resistances. (6 marks)



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENC
(AEROSPACE ENGINEERING), BACHELOR OF SCIENCE (AGRICULTURAL &
BIOSYSTEMS ENGINEERING), BACHELOR OF SCIENCE (CIVIL ENGINEERING),
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONIC ENGINEERING), BACHELOR
OF SCIENCE (MECHANICAL ENGINEERING) AND BACHELOR OF SCIENCE
(PETROLEUM ENGINEERING)

ECU 203: ENGINEERING MATHEMATICS VIII (TRANSFORM METHODS)

2ND SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS: Answer QUESTION ONE and any other TWO questions.

QUESTION ONE

a) Compute the following integrals using Gamma (Γt) and Beta $B(m,n)$ special functions

i) $\int_0^{\infty} x^3 e^{-x} dx$ (2marks)

ii) $\int_0^1 x^2 (x - x^2)^5 dx$ (3marks)

b) Use power series method to solve the differential equation $y'' - xy = 0$ about $x_0 = 0$.

(5marks)

c i) Define the term Laplace Transform.

(2 marks)

ii) Using the definition stated in a (i) above, find the Laplace Transform of $f(t) = t^2 + e^{\frac{1}{2}t}$

(3marks)

d) Find the Inverse Laplace Transform of $F(s) = \frac{s-2}{s^2-4s+29}$

(5marks)

e) Obtain a Fourier series for $f(x) = 1 - x^2$ when $-1 \leq x \leq 1$.

(5marks)

f) An alternating current i after passing through a rectifier has the form

$$i = \begin{cases} I_0 \sin x & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } \pi \leq x \leq 2\pi \end{cases}$$

Where I_0 is the maximum current and the period is 2π . Express the function i as a Fourier series. (5marks)

QUESTION TWO

(a) Use the Laplace Transform table to find the Laplace Transform of the functions

(2marks)

(i) $f(t) = e^{\frac{1}{2}t} \sin \frac{1}{3}t$

(2marks)

(ii) $f(t) = e^{-3t} t^2$

(6marks)

(b) Find the inverse Laplace transform $f(t)$ given that $F(s) = \frac{3s^2 - 2s - 1}{(s-3)(s^2+1)}$

(c) Solve the following Initial Value Problems using Laplace Transform methods

(5marks)

(i) $y'' - 3y' + 2y = 6e^{-t}$ $y(0)=3$ and $y'(0)=3$

(5marks)

(ii) $y'' - y = 1$ $y(0)=1$ and $y'(0)=2$

QUESTION THREE

3a) Determine the singular points of the differential equation below and classify them as regular or irregular

$$x(x^2 - 4)y'' + (x + 2)y' + 3y = 0$$

(4marks)

b) Using Rodrigue's formula for the n^{th} order Legendre's polynomial

$$P_n(x) = \frac{1}{2^n n! dx^n} (x^2 - 1)^n$$

find a polynomial solution to the Legendre equation $(1 - x^2)y'' - 2xy' + 12y = 0$. (5marks)

c) Determine the recurrence relation of the 2^{nd} order differential equation $y'' - xy' + 2y = 0$ near ordinary point $x_0 = 0$ using power series. (6marks)

d i) The second order equation $x^2 y'' + xy' + (x^2 - p^2)y = 0$ where p is a constant is called the Bessel's differential equation. Show that the equation has a regular singular point at $x = 0$.

(2marks)

ii) The Bessel function of the first kind of order p is denoted by

$J_p(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(p+r+1)} \left(\frac{x}{2}\right)^{2r+p}$ and gamma $\Gamma(r+1) = r!$. Prove that the Bessel's function of order zero is given by $J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$

(3marks)

QUESTION FOUR

a i) Write down the representation of the Fourier series of a function $f(x)$. (1mark)

ii) Determine whether each of the following functions is odd, even or neither odd nor even and hence identify the nature of their Fourier series expansion.

i) $x \sin x$

ii) $\frac{x}{x^2-1}$

iii) $x^3 - x^2 - 1$

(6marks)

b) Consider the periodic function $f(x) = \begin{cases} 2x & 0 \leq x < 3 \\ 0 & -3 < x < 0 \end{cases}$, period 6. Graph the periodic function and obtain a Fourier series for the function. (6marks)

c) Obtain a Fourier series for $f(x) = x + x^2$ when $-\pi \leq x \leq \pi$. Hence show that

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12} \quad (7marks)$$

QUESTION FIVE

a) The Beta function is denoted by $B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$ prove that $\frac{B(m+1, n)}{B(m, n)} = \frac{m}{m+n}$ (3marks)

b) Using special functions compute

i) $\Gamma\left(\frac{5}{2}\right) B\left(\frac{3}{2}, 1\right)$

(3marks)

ii) $\int_0^{\frac{\pi}{2}} \cos^{\frac{5}{2}} \theta \sin^3 \theta \, d\theta$

(3marks)

iii) $\int_0^1 x^5 (1-x)^3 \, dx$

(2marks)

iv) $\int_0^{\infty} x^2 e^{-x^2} \, dx$

(3marks)

c) Prove that $\int_0^3 \frac{dx}{\sqrt{3x-x^2}} = \pi$ by using gamma and beta functions.

(4marks)

d) State two applications of Gamma and Beta functions in Engineering.

(2marks)

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$	2. e^{at}	$\frac{1}{s-a}$
3. $t^n, n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$	4. $t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6. $t^{n-\frac{1}{2}}, n = 1, 2, 3, \dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1) \sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7. $\sin(at)$	$\frac{a}{s^2 + a^2}$	8. $\cos(at)$	$\frac{s}{s^2 + a^2}$
9. $t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$	10. $t \cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11. $\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$	12. $\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13. $\cos(at) - at \sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$	14. $\cos(at) + at \sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$
15. $\sin(at + b)$	$\frac{s \sin(b) + a \cos(b)}{s^2 + a^2}$	16. $\cos(at + b)$	$\frac{s \cos(b) - a \sin(b)}{s^2 + a^2}$
17. $\sinh(at)$	$\frac{a}{s^2 - a^2}$	18. $\cosh(at)$	$\frac{s}{s^2 - a^2}$
19. $e^{at} \sin(bt)$	$\frac{b}{(s-a)^2 + b^2}$	20. $e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2 + b^2}$
21. $e^{at} \sinh(bt)$	$\frac{b}{(s-a)^2 - b^2}$	22. $e^{at} \cosh(bt)$	$\frac{s-a}{(s-a)^2 - b^2}$
23. $t^n e^{at}, n = 1, 2, 3, \dots$	$\frac{n!}{(s-a)^{n+1}}$	24. $f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25. $u_c(t) = u(t-c)$	$\frac{e^{-cs}}{s}$	26. $\delta(t-c)$	e^{-cs}
27. $u_c(t)f(t-c)$	$e^{-cs}F(s)$	28. $u_c(t)g(t)$	$e^{-cs}\mathcal{L}\{g(t+c)\}$
29. $e^{ct}f(t)$	$F(s-c)$	30. $t^n f(t), n = 1, 2, 3, \dots$	$(-1)^n F^{(n)}(s)$
31. $\frac{1}{t}f(t)$	$\int_s^\infty F(u) du$	32. $\int_0^t f(v) dv$	$\frac{F(s)}{s}$
33. $\int_0^t f(t-\tau)g(\tau) d\tau$	$F(s)G(s)$	34. $f(t+T) = f(t)$	$\frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$
35. $f'(t)$	$sF(s) - f(0)$	36. $f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
37. $f^{(n)}(t)$	$s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$		

Table of Laplace Transforms

Table Notes

1. This list is not a complete listing of Laplace transforms and only contains some of the more commonly used Laplace transforms and formulas.

Recall the definition of hyperbolic functions.

$$\cosh(t) = \frac{e^t + e^{-t}}{2} \quad \sinh(t) = \frac{e^t - e^{-t}}{2}$$

2. Be careful when using "normal" trig function vs. hyperbolic functions. The only difference in the formulas is the "+a²" for the "normal" trig functions becomes a "-a²" for the hyperbolic functions!
3. Formula #4 uses the Gamma function which is defined as

$$\Gamma(t) = \int_0^{\infty} e^{-x} x^{t-1} dx$$

If n is a positive integer then,

$$\Gamma(n+1) = n!$$

The Gamma function is an extension of the normal factorial function. Here are a couple of quick facts for the Gamma function

$$\begin{aligned} \Gamma(p+1) &= p\Gamma(p) \\ p(p+1)(p+2)\cdots(p+n-1) &= \frac{\Gamma(p+n)}{\Gamma(p)} \\ \Gamma\left(\frac{1}{2}\right) &= \sqrt{\pi} \end{aligned}$$

ECU 203 TEACHING SCHEDULE

SEMESTER TWO 2021/2022

WEEK	TOPIC AND CONTENT
1	SPECIAL FUNCTIONS: Gamma functions and Beta functions
2	Bessel functions and Legendre polynomials
3	Applications of special functions in solving Engineering problems
4	LAPLACE TRANSFORMS: Definition and notation
5	Transforms of powers, exponential and trigonometric functions
6	Basic properties, shift and factor rules Continuous Assessment Test (C.A.T.) ONE
7	Inverse Transforms, Heaviside and Delta functions. Application of Laplace transforms: convolution theorem, solution of ordinary differential equations (O.D.Es)
8	FOURIER SERIES: Periodic functions: sine and cosine functions
9	Determination of coefficients for odd and even functions
10	Properties of Fourier transforms (FT) and Transform rules
11	Convolution theorem, delta and Dirac functions
12	Applications: solution differential equations by Fourier transform methods
13	Review of Semester's work. C.A.T.TWO
14	END OF SEMESTER EXAMINATIONS
15	END OF SEMESTER EXAMINATIONS

ECU 203: ENGINEERING MATHEMATICS VIII

Prerequisites: ECU 203: Engineering Mathematics VIII

Purpose

The purpose of the course is to impart knowledge on solution and applications of First and higher order differential equations.

Expected outcomes

At the end of the course, a student should

1. Be able to solve Laplace Transforms
2. Be able to apply Laplace Transforms to solve differential equations
3. Be able to solve higher order linear differential equations and apply these Laplace and Fourier Transforms.

Course content

Laplace Transform: Definitions and notation; transforms of powers, exponential and trigonometric functions; scales, shift and factors rules; inverse transforms, application to Heaviside and Delta functions and solution of differential equations; Transfer functions, convolution theorem and discrete systems: Fourier series: Periodic functions, including sine and cosine series, determination of coefficients, even and odd functions: Fourier Transform (FT) and non-periodic functions: Properties of FT and Transform rules including on differentiation, convolution theorem, delta and Dirac functions: Solutions of Differential Equations by Laplace Transform and Fourier: Special functions: Bessel functions, Legendre Polynomials; Gamma functions.

Teaching Organization

Lectures: 2 hours per week; Tutorials: 2 hours per week.

Assessment

Ordinary Examination at end of Semester: 70 % Continuous Assessment:
30 % where 20 % shall be continuous assessment tests and 10 % shall be assignments.

Textbooks and References

- Jordan D. W., and Smith P., Mathematical Techniques, An introduction for the engineering, physical and mathematical sciences; Second edition, Oxford University Press, 1997.
- Kaplan W., Advanced Calculus, Second Edition, Addison-Wesley Publishing Company, 1973.
- K.A. Stroud and Dexter J. Booth, Advanced Engineering Mathematics Industrial Park Inc, New York, 2011

QUESTION ONE

a) Given that gamma t is defined as $\Gamma t = \int_0^{\infty} x^{t-1} e^{-x} dx$ for $t > 0$. Prove that $\Gamma(t+1) = t\Gamma t$.
 Hence compute $\int_0^{\infty} x^5 e^{-x} dx$. (5marks)

b) Use gamma and beta functions to compute

i) $\frac{\Gamma 3 \Gamma 0.5}{\Gamma 5.5}$ ii) $B(6, 4)$ iii) $\int_0^1 x^3 (1-x)^4 dx$ iv) $\int_0^{\frac{\pi}{2}} \sin^5 x \cos^7 x dx$. (8marks)

c) Given that $\Gamma \frac{1}{2} = \sqrt{\pi}$ use a suitable substitution to show that $\int_0^2 x \sqrt[3]{8-x^3} dx = \frac{16\pi}{9\sqrt{3}}$. (4marks)

QUESTION TWO

a) Explain the difference between an ordinary point and a singular point in relation to a linear second order ordinary differential equation.

$y'' + P(x)y' + Q(x)y = 0$ [2marks]

b) Locate and classify the singular points of the differential equations

$(x^2 - 1) \frac{d^2 y}{dx^2} + (x^3 - x^2) \frac{dy}{dx} + (1 - x)y = 0$ [4marks]

c) The Bessel equation of order p is of the form

$x^2 y'' + xy' + (x^2 - p^2)y = 0$. Give an example of the Bessel equation and show that $x=0$ is a regular singular point of the equation. [4marks]

QUESTION THREE

a) Solve the differential equation below using an appropriate series method. [5marks]

$y'' - 4y = 0$.

b) Find a particular polynomial solution to the Legendre equation

$(1 - x^2)y'' - 2xy' + 6y = 0$ (Series solution not required). [3marks]

c) Use power series method to solve the Legendre differential equation at $x = 0$.

$(1 - x^2)y'' - 2xy' + n(n+1)y = 0$ (Attach the solution) [10marks]

ECU 203: ENGINEERING MATHEMATICS VIII -CAT II-10TH MARCH 2023

ATTEMPT ALL THE QUESTIONS

- a) Define the Laplace transform and hence determine the Laplace transform of the function

$$f(t) = t(t - 1) - \sin 2t + e^{3t} \quad (5 \text{ marks})$$

- b) A first-order differential equation involving current i in a series R-L circuit is given by:

$$\frac{di}{dt} + 5i = \frac{E}{2} \text{ and } i = 0 \text{ at time } t = 0. \text{ Use Laplace transforms to solve for } i \text{ when}$$

$$E = 50 \sin 5t \quad (5 \text{ marks})$$

- c) Use Laplace transform tables to determine the Laplace transform of $f(t) = 6e^{-5t} + e^{3t} +$

$$5t^3 - 9 \quad (5 \text{ marks})$$

- d) Given that $F(s) = \frac{6s^2 + 62s + 92}{(s+1)(s^2 + 10s + 21)}$, determine $f(t)$ (5 marks)

- e) Use Laplace transform to solve the initial value problem $y'' + 3y' + 2y = 6e^{-t}$, given that

$$y(0) = 1 \text{ and } y'(0) = 2 \quad (10 \text{ marks})$$

DEFINITIONS AND FORMULAE

Indefinite integrals of common functions

[In the following we take $a > 0$ and omit the additive constant.]

$\frac{f(x)}{x^n (n \neq -1)}$	$\frac{\int f(x) dx}{x^{n+1}/(n+1)}$
$1/x$	$\ln x \text{ if } x > 0, \ln(-x) \text{ if } x < 0$ (i.e. $\ln x , x \neq 0$)

$\frac{1}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$
-----------------------	-------------------------------------

$\frac{1}{x^2 - a^2}$	$\frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $
-----------------------	---

$\frac{1}{\sqrt{(x^2 + a^2)}}$	$\sinh^{-1} \frac{x}{a}$
--------------------------------	--------------------------

$\frac{1}{\sqrt{(x^2 - a^2)}}$	$\cosh^{-1} \frac{x}{a} \text{ if } x > a, -\cosh^{-1} \left(\frac{-x}{a} \right) \text{ if } x < -a$
--------------------------------	--

$\frac{1}{\sqrt{(a^2 - x^2)}}$	$\sin^{-1} \frac{x}{a}$
--------------------------------	-------------------------

$\sin x$	$-\cos x$
----------	-----------

$\cos x$	$\sin x$
----------	----------

$\tan x$	$\ln \sec x $
----------	----------------

$\cot x$	$\ln \sin x $
----------	----------------

$\sec x$	$\ln \sec x + \tan x = \ln \left \tan \left(\frac{1}{2}x + \frac{1}{2}\pi \right) \right $
----------	--

$\csc x$	$\ln \left \tan \frac{1}{2}x \right $
----------	--

$e^{ax} \sin bx$	$\frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$
------------------	--

$e^{ax} \cos bx$	$\frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$
------------------	--

$\sin^2 x$	$\frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right)$
------------	--

$\cos^2 x$	$\frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right)$
------------	--

$\sinh x$	$\cosh x$
-----------	-----------

$\cosh x$	$\sinh x$
-----------	-----------

For logarithmic forms of inverse hyperbolic functions see p. 3.

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int \frac{du}{dx} v dx.$$

Reduction formulae for trigonometric integrals

$$\int_0^{\frac{1}{2}\pi} \sin^m x dx = \frac{m-1}{m} \int_0^{\frac{1}{2}\pi} \sin^{m-2} x dx; \quad \int_0^{\frac{1}{2}\pi} \cos^m x dx = \frac{m-1}{m} \int_0^{\frac{1}{2}\pi} \cos^{m-2} x dx;$$

$$\int_0^{\frac{1}{2}\pi} \sin^m x \cos^n x dx = \frac{m-1}{m+n} \int_0^{\frac{1}{2}\pi} \sin^{m-2} x \cos^n x dx = \frac{n-1}{m+n} \int_0^{\frac{1}{2}\pi} \sin^m x \cos^{n-2} x dx.$$

[These results hold provided that the exponents in the reduced form are greater than -1 . There are analogous reduction formulae with other intervals of integration ($\frac{1}{2}k_1\pi, \frac{1}{2}k_2\pi$) with k_1, k_2 integral.]



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL AND ELECTRONIC ENGINEERING
EEE 207: DATABASE MANAGEMENT

2ND SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS:

- Q1. (a) Identify TWO (2) characteristics of Graph Databases not found in SQL databases. (2 marks)
- (b) Give an example of the following using relational algebra notations:
- i. Natural join (2 marks)
 - ii. Union (2 marks)
 - iii. Projection (2 marks)
- (c) Using an example, enumerate three different types of SQL, DDL statements used in relational databases. (6 marks)
- (d) Identify the key consideration when selecting a primary key in relational database. (3 marks)
- (e) Identify two modeling techniques used for designing relational database structures. (2 marks)
- (f) Using ER diagram in either ONE OF Chen, crow foot or UML notation, identify the symbols for:
- i. Entity (1 mark)
 - ii. Primary key (1 mark)
 - iii. Relation (1 mark)
 - iv. Foreign Key (1 mark)
- (g) List the names of three types of indexes. (3 marks)

- (h) JSON is becoming a popular data exchange protocols format, explain what the acronym stands for and describe three key characteristics. (4 marks)
- Q2. (a) Consider the following expressions and for each expression, explain in words what the expression does and the meaning of each symbol or notation. (4 marks)
- $\delta_{dept\ name = "EEE"} (instructor)$
 - $\Pi_{student.ID, student.name, takes.courseid, takes.coursename} (takes |X| student)$ (4 marks)
- (b) Identify an example of an SQL statement that performs EACH ONE of the CRUD operations. (4 marks)
- (c) The DML statements allow you to aggregate data. Write an SQL statement using the following table 'instructor (id, instructor name, salary, department, date joined)' and aggregate the data to show.
- the total salaries for instructors in each department. (4 marks)
 - the total number of instructors in each department. (4 marks)
- Q3. (a) The following details are maintained about the books which include ISBN number used to identify the book uniquely, book title, author(s), publisher and copy number. The following details are maintained about the library members include: member no, name, date of birth, telephone and email. The borrowing record contains the member no, ISBN number, copy number, borrow date, return due date, actual return date and days borrowed. Using this information:
- List all the possible entities (5 marks)
 - Identify the primary key for each entity and indicate their type. (5 marks)
 - Write statement describing the relationship between any two entities. (4 marks)
 - Using the Crow Foot Notation ERD Modeling technique draw, the two entities identified above, the relationship between them, their attributes and any keys. (6 marks)
- Q4. (a) Identify the component in a relational database that manages indexes. (2 marks)

- (b) Describe four key characteristics of the following types of indexes
- i. Ordered indexes (4 marks)
 - ii. B+Tree (4 marks)
- (c) A database has 40 million records and have been indexed using a hash index on a disk storage with an average seek time of 10 milliseconds and a block size of 4 kilobytes:
- i. explain the time complexity of the static hash index. (4 marks)
 - ii. estimated the time it will take to access a record in the database using this type of index. (6 marks)

- Q5. (a) Describe the acronym ACID in the context of database transaction management. (4 marks)
- (b) Describe the following Data Control Language statements in SQL.
- i. Grant (2 marks)
 - ii. Revoke (2 marks)
- (c) Using an SQL example, discuss the purpose of each for the following TCL statements:
- i. Commit (3 marks)
 - ii. Rollback (3 marks)
- (d) Describe the purpose of each of the ETL stages in data warehousing. (6 marks)



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 204: CIRCUIT THEORY II

2ND SEMESTER 2022/2023

TIME: 2 Hours

INSTRUCTIONS:

Answer question ONE and any other TWO questions.

Question ONE is compulsory and carries 30 Marks.

The remaining questions carry 20 Marks

QUESTION ONE

- a) Explain in three phase circuits
- i) Why a fuse is never placed on a neutral wire (2 marks)
 - ii) The disadvantages of low power factor (3 marks)
 - iii) The importance of phase sequence in 3 phase circuits (2 marks)
 - iv) What happens when one phase of Δ connected alternator is reversed (3 marks)
- b) Explain in a three phase circuit
- i) How phase and line circuits are related (2 marks)
 - ii) When it is appropriate to use star/delta transformation (3 marks)
 - (iii) The importance of phase sequence (3 marks)
- c) The load to a 3-phase supply comprises three similar coils connected in star.
The line currents are 25A and the kVA and kW inputs are 20 and 11 respectively.
Find:
- (i) The phase and line voltages (2 marks)
 - (ii) The KVA input; and (2 marks)
 - (iii) Resistance and reactance of each coil (2 marks)

QUESTION TWO

- a) Explain the following terms in a magnetic circuit
- (ii) Leakage flux (2 marks)
 - (iii) Coercive force (2 marks)
 - (iv) Reluctance (1 mark)
- b) Discuss:
- (i) The B-H curve of a ferro- magnetic material (3 marks)
 - (ii) The similarities of a magnetic and Electric Circuits (4 marks)
 - (iii) The Industrial applications of Hysteresis loop (2 marks)
 - (iv) If a magnetic circuit consumes energy (2 marks)
- c) An iron ring of cross sectional area 6 cm^2 is wound with a wire of 200 turns and has a saw cut of 4mm. Calculate the magnetizing current required to produce a flux of 0.2 mwb if mean length of magnetic path is 40cm and relative permeability of iron is 500. (4 marks)

QUESTION THREE

- a)
- i) Explain the term resonance (2 marks)
 - ii) What is the effect of impedance at resonance (2 marks)
 - iii) The Industrial applications of resonance (2 mark)
- b)
- i) How is the band width affected by factor Q factor of the Circuit (2 marks)
 - ii) Discuss resonance in R-L-C Circuit (3 marks)
 - iii) Discuss physical meaning of Q-factor (3 marks)
- c) A circuit having resistance of 4Ω and inductance of 0.5H and variable capacitance in series is connected across 100V 50HZ Supply. Calculate:
- i) Capacitance to give resonance (2 marks)
 - ii) Voltage across inductance and capacitance at resonance (2 marks)
 - iv) The Q – factor of the Circuits (2 marks)

QUESTION FOUR

- a) State the following Laws of Electromagnetic Induction
- i) Fleming's Right Hand Rule (2 marks)
 - ii) Faraday's Laws (2 marks)
 - iii) Lenz's Law (2 marks)
- b) Explain briefly
- (i) The co-efficient of Coupling (3 marks)
 - (ii) Time Constant and its Units (3 marks)
 - (iii) Two magnetically coupled coils have self-inductances, L_1 and $L_2 = 400\text{mH}$. if the co-efficient of coupling is 0.8, find the value of mutual inductance between the coils. What would be the maximum possible mutual inductances? (3 marks)
- c) When two Coils are connected in series, there effective inductance is found to be 10H , when connections of one Coil is reversed the effective inductance is 6H . If the coefficient of coupling is 0.6. Calculate the self-inductance of each coil and the mutual inductance. (5 marks)

QUESTION FIVE

- a) Explain briefly the following:
- i) The significance time constants and its units. (2 marks)
 - ii) The transfer function (2 marks)
 - iii) Decay of current in an Inductive circuit (3 marks)
- b) Explain briefly
- i) Software Proteus and how it is used to solve circuit problems (4 marks)
 - ii) Why there is no fuse on the neutral wire of a.c circuits? (3 marks)
- c) A coil has a time constant 0.1s and an inductance of 10H . Find:
- i) The value of current 0.1s after switching to steady p.d of 100V (3 marks)
 - ii) The time taken for the current to reach half its steady state value (3 marks)



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 210: AUTOCAD

2ND SEMESTER 2022/2023

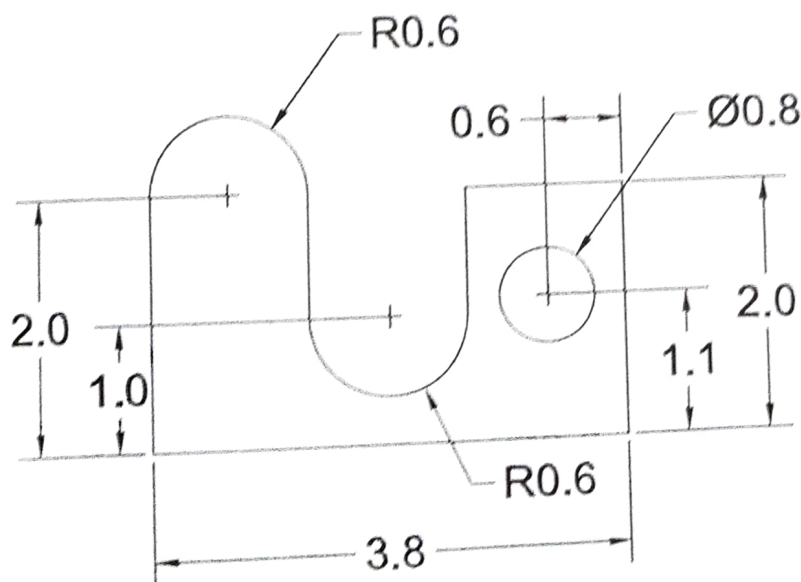
TIME: 2 Hours

INSTRUCTIONS:

Answer Question 1 and Any Other TWO Questions

QUESTION 1

- a) Explain in details the process of drawing the diagram below complete with the relevant shapes and tools used? (5Mks)



- b) If no suitable layer exists, you create one in the Layer Properties Manager palette. Detail the steps that are followed in setting up a new layer? (10Mks)
- c) Analysis of AutoCAD users' actions suggests that one of the most commonly used functions is undo. Give the five approaches that can be used to undo a command: (5Mks)
- d) The mouse is a very important tool to manipulate and work around AutoCAD. Draw the image of a mouse and name the various buttons and their functions? (6Mks)

- e) Highlight how to save a drawing and the difference between using the 'Save' command and the 'Save As' command (4Mks)

QUESTION 2

- a) Detail the steps in executing the Move command: (7Mks)
- b) Describe a block in AutoCAD and how it is created? (5Mks)
- c) Explain four benefits of using blocks: (8Mks)

QUESTION 3

- a) When you first start a drawing, you need to decide what the length of one unit represents—an inch, a foot, a centimeter, a kilometer, or some other unit of length. Discuss how to adjust the unit settings and the various parameters that are adjusted (5Mks)
- b) When drafting, give the shortcut keys or actions for the following? (5Mks)
- To open **Help** for information about the command in progress
 - To **repeat** the previous command
 - To see various **options**, the two possible keys
 - To **cancel** a command in progress or if you ever feel stuck
- c) After you create layers and draw objects on them, you can manipulate the layers in various ways. Give the three ways that can be used to manipulate the layers? (6Mks)
- d) AutoCAD operates in two parallel universes, called *spaces*, which AutoCAD indicates with either a status bar button or two or more tabs in the lower left section of the drawing area. Describe these two spaces: (4Mks)

QUESTION 4:

- a) Describe the steps of setting up a drawing for printing in layout on an A3 paper and scale the drawing to 1:50. (10Mks)
- b) Describe how you can draw a circle of radius 500mm and mirror it, retain the original circle? (5Mks)

- c) Describe the 'PAN' command and how to use it, hence differentiate it from the MOVE command? (5Mks)

QUESTION 5

- a) Explain how 'Trim' and 'Extend' work and how they are executed? (6Mks)
- b) The Zoom command has 11 options. The most important ones are five. Describe these five: (10Mks)
- c) Explain how to change the background colour of the working area in AutoCAD? (4Mks)